JMP TEST 2 · FRONT

TWO-SAMPLE T · MATCHED PAIRS | ▼ RED TRIANGLE · › THEN CLICK · **GREEN = WRITE IT, FILL THE BLANKS**

SYMBOLS

$\bar{x}_1$ $\bar{x}_2$ the two sample averages
s_1 s_2 the two sample SDs
n_1 n_2 the two sample sizes
F ANOVA test statistic; big F = real effect
q studentized range — Tukey uses it
Prob > t one-sided p, upper tail
μ_1 μ_2 "mew" true means, unknown
s_p pooled SD — one SD for both groups
α "alpha" the bar you compare p to
MSE Mean Square Error = s_p^2
HSD Tukey cutoff; |diff| > HSD = different
Prob > F the ANOVA p-value
$\bar{d}$ "d-bar" average of the differences
SE standard error = spread of the estimate
H_o the null — always has an =
RMSE √MSE, the typical error
LSMean least squares mean = a cell average
A*B the interaction term
μ_d true mean difference; H_o: $\mu_d = 0$
df 2-sample n_1+n_2-2 · paired n−1
H_1 the alternative: <, > or ≠
k how many groups
Prob > |t| the **two-sided** p-value
t* critical t — needs df

WHICH TEST? — decide this first, every time

Count the variables, then ask who the subjects are

```
2 GROUPS different subjects → two-sample t › Fit Y by X
2 GROUPS same subject twice → Matched Pairs
1 FACTOR 3+ groups → one-way ANOVA › Fit Y by X
2 FACTORS → two-way ANOVA › Fit Model › Cross
```

Tie-breaker: could you shuffle one column without losing info? Yes → two-sample. No → matched pairs.

Data layout is OPPOSITE for those two

```
FIT Y BY X stacked — 1 response column + 1 group column
MATCHED PAIRS side by side — 2 response columns
```

Column won't go in the X box? Right-click header › Column Info › Modeling Type → **Nominal**. Text won't type in? Data Type → **Character**.

T1 · TWO INDEPENDENT MEANS — two-sample t

a Raw data, or just summary numbers?

```
RAW ROWS Analyze › Fit Y by X › Y = response, X = group › OK
ONLY N, X̄, S Help › Sample Index › Tool › Calculators ›
TestTwoSampleMeans › t-test › Assume Equal Variances › pick H1
```

Fit Y by X cannot run on summary statistics. It needs actual rows. **The calculator gives you NO Brown-Forsythe and NO quantile plot** — argue conditions from the problem text and the SD ratio by hand.

b Turn on everything you need, once

```
▼ Means/Anova/Pooled t · t Test · Unequal Variances · Normal Quantile Plot
```

That one click-set answers conditions, test statistic, p-value and CI.

c Variables and parameters

```
SAY The explanatory variable is _____ , which is categorical with 2 levels.
The response is _____ , which is quantitative.
```

```
SAY μ1 = the true mean _____ for all _____ , and μ2 = the true mean for all _____ .
```

```
SAY The point estimate of the difference is x̄1 − x̄2 = _____ − _____ = _____ .
```

Parameters are about the whole population, never the numbers you were handed.

d Hypotheses — watch the wording

```
TWO-SIDED "a difference", "differ", "changed" → H1: μ1 ≠ μ2
ONE-SIDED "higher", "greater", "more than", "improved" → H1: μ1 > μ2
```

```
SAY Ho: μ1 = μ2  H1: μ1 _____ μ2
```

e Conditions — all three, in this order

```
SAY Independence: subjects were randomly assigned (or randomly sampled) and each appears in only one group, so the two samples are independent.
```

```
SAY Normality: n = _____ per group. If n ≥ 30: the Central Limit Theorem applies, so no plot is needed. If n < 30: the Normal Quantile Plot shows the points following their lines with no curve or outliers.
```

```
SAY Equal variances: the SDs are _____ and _____ , a ratio of _____ which is under 2, and Brown-Forsythe gives p = _____ > 0.05, so pooling is appropriate.
```

SD ratio under **2** = variance ratio under **4**. Fails? Don't stop — quote the **t Test** (Welch) box instead of Pooled t. Conditions 1 and 2 are required either way.

f Test statistic and p-value

```
EQUAL VAR Pooled t Test box → t Ratio, DF = n1+n2−2, Prob > |t|
UNEQUAL t Test box ("Assuming unequal variances") → decimal DF
```

```
SAY If Ho were true, there would be about a _____ chance of getting a difference in sample means this far from 0 just by random chance.
```

One-sided? Use **Prob > t** or **Prob < t**, never Prob > |t|. Read the header (e.g. *B–A*) to see the subtraction order. Safety check: the right one-sided p is **below 0.5** when the data lean the way H_1 says. Got 0.9997? Take the other row.

g Conclusion

```
SAY Since p = _____ is less than α = _____ , we reject Ho. There is sufficient evidence to say that _____ .
```

```
SAY Since p = _____ is greater than α = _____ , we fail to reject Ho. There is not sufficient evidence to say that _____ .
```

Never "accept H_o". Never "prove".

h Confidence interval

```
READ Upper CL Dif and Lower CL Dif in the same t-test box
```

```
SAY We are 95% confident the true difference in mean _____ is between _____ and _____ .
```

CI excludes 0 ↔ you rejected H_o. They always agree. If asked "will it contain 0?" before computing → answer from the p-value.

T2 · MATCHED PAIRS — same subject measured twice

a Run it

```
PATH Analyze › Specialized Modeling › Matched Pairs
THEN put BOTH columns into Y, Paired Response › OK
```

One box, two columns. Do **not** stack this data.

b Equivalent route, if asked the long way

```
MAKE COL Cols › New Column › Column Properties › Formula → after − before
THEN Analyze › Distribution › ▼ › Test Mean › 0
```

Identical t and p. The difference column is also how you check normality.

c Hypotheses

```
SAY Let μd be the true mean difference ( _____ minus _____ ) for all _____ .
```

```
SAY Ho: μd = 0  H1: μd _____ 0
```

d Conditions — only two, and one is different

```
SAY Independence: the pairs were randomly sampled and one pair does not affect another. The two values within a pair are dependent by design, which is the point.
```

```
SAY Normality: the differences are approximately normal — the quantile plot of the difference column shows no curve or outliers.
```

Normality applies to the **differences**, not to either original column. There is **no** equal-variance condition here.

e Read it and conclude

```
READ Mean Difference · t-Ratio · DF = n−1 · Prob > |t| · Upper95% / Lower95%
```

5 pairs → **df = 4**, not 8. JMP prints the subtraction order in the label (e.g. *2022-2020*); the second column minus the first flips only the **sign**, never the p-value.

```
SAY Since p = _____ < α = _____ , we reject Ho. There is sufficient evidence of a difference in mean _____ between _____ and _____ .
```

f Confidence interval for the difference

```
READ Upper95% and Lower95% in the same box
```

```
SAY We are 95% confident the true mean difference in _____ is between _____ and _____ . The interval excludes 0, which agrees with rejecting Ho.
```

g Why pairing is worth it

Pairing strips out subject-to-subject variation, so t gets bigger. Run wrongly as two independent samples the same data gives a much weaker result. That's the answer if asked why you paired.

REVIEW SESSION 2 · CHECK YOUR WORK

Snap Beans — *two-sample t, summary stats, one-sided, α=0.01*

```
PATH Calculators › TestTwoSampleMeans › Assume Equal Variances
ANSWER sp=27.61 · SE=6.903 · t=5.65 · df=62 · p<0.0001 → reject
```

SD ratio 1.2 ✓ · n=32 so CLT covers normality ✓

Sports Attendance — *matched pairs, α=0.05*

```
ANSWER d̄=176,192.8 · sd=34,838.6 · t=11.31 · df=4 · p=0.0003 → reject
95% CI (132,935 , 219,451) — excludes 0 ✓
```

As independent samples it drops to t=5.53. Pairing doubled it.

Fruit Flies — *one-way ANOVA, 5 groups of 25*

```
DF CHECK Treatment 4 · Error 120 · C. Total 124
```

Handout says "four treatments" but the factor has **5** levels. F large, p tiny → Tukey.

Battery Life — *two-way, 3×3, n=4 per cell*

```
DF CHECK Model 8 · Error 27 · C. Total 35
EFFECT TESTS Material F=7.91 p=0.0020 · Temp F=28.97 p<.0001 ·
Material*Temp F=3.56 p=0.0186
```

Interaction **IS** significant → compare all 9 cells, not the main effects. MSE=675.21, HSD=61.82.

```
CIRCLE A B15 155.75 · C70 145.75 · C15 144.00 · A15 134.75 · B70 119.75
CIRCLE B C70 · C15 · A15 · B70 · C125 85.50
CIRCLE C C125 85.50 · A125 57.50 · A70 57.25 · B125 49.50
```

C125 is in two circles: matches B70 above and A125 below, yet those two differ. **Material C holds up best → recommend C.**

JMP TEST 2 · BACK

ONE-WAY ANOVA · TWO-WAY ANOVA · TUKEY | ▼ RED TRIANGLE · **GREEN = WRITE IT, FILL THE BLANKS**

T3 · ONE-WAY ANOVA — one factor, 3+ groups

a Run it

`PATH` Analyze › Fit Y by X › Y = response, X = factor › OK
▼ Means/Anova · Unequal Variances · Normal Quantile Plot

X must be **Nominal** (red bars). If it's Continuous you get a regression, not an ANOVA.

b Hypotheses

`SAY` H_0: $\mu_1 = \mu_2 = \mu_3 = \ldots$ (all group means are equal)

`SAY` H_1: **at least one** mean differs from the others

Never write "all means are different". That is wrong and gets marked.

c DF check before you read anything

`CHECK` Factor DF = k−1 · Error DF = N−k · C. Total = N−1

5 groups of 25: 4, 120, 124. Wrong DF → wrong column selected or wrong modeling type.

d Conditions

`SAY` **Independence:** subjects were randomly assigned to the groups.

`SAY` **Normality:** n = _____ per group and the quantile plots show points near their lines.

`SAY` **Equal variances:** largest SD / smallest SD = _____ , under 2, and Brown-Forsythe p = _____ > 0.05.

With 3+ groups the rule is **largest over smallest**, not pairwise.

e The F test and your conclusion

`READ` Analysis of Variance table → **F Ratio** and **Prob > F**

`SAY` Since p = _____ < α = _____ , we reject H_0. There is sufficient evidence that at least one mean _____ differs.

`SAY` Since p = _____ > α = _____ , we fail to reject H_0. There is not sufficient evidence that the means differ, so we do **not** make multiple comparisons.

f Multiple comparisons — only if F was significant

▼ Compare Means › **All Pairs, Tukey HSD**

"If appropriate" means exactly this: F not significant → stop. Why not 10 t-tests? 10 tests at 5% each = ~40% chance of a false positive; Tukey holds the **overall** rate at 5%.

g Reading connecting letters

`RULE` share a letter → **NOT** different · no letter in common → **different**

`SAY` Tukey HSD comparisons at an overall 0.05 level showed that _____ did not differ significantly, while _____ was significantly _____ than _____ .

T4 · TWO-WAY ANOVA — two factors, Fit Model

a Name the factors, their levels, and the response

`SAY` The two factors are _____ , which is **categorical** with _____ levels (_____), and _____ , which is **categorical** with _____ levels (_____).

`SAY` The response variable is _____ , which is **quantitative**.

`SAY` There are _____ treatment combinations with _____ units each, so the design is balanced and the interaction can be tested.

b Run it — the Cross step is everything

1 Analyze › **Fit Model** › Y, Response = the response
2 Add **both** factors
3 select **BOTH** inside Construct Model Effects › click **Cross**
4 Run

Three lines must show before Run: *A*, *B*, *A*B*. Only two → you skipped Cross and the interaction is missing. (*Macros › Full Factorial* does all of it in one click.)

c DF check — your safety net

`FORMULA` Model DF = (a−1) + (b−1) + (a−1)(b−1)
3×3, N=4 Model **8** · Error **27** · C. Total **35**
3×2, N=5 Model **5** · Error **24** · C. Total **29**

Model DF too small → no interaction in the model. Go back and Cross. This check catches the wrong-file mistake too.

d Three sets of hypotheses — write all three

`SAY` **Interaction** — H_0: there is no interaction between _____ and _____ . H_1: there is an interaction.

`SAY` **Factor A** — H_0: $\mu_1 = \mu_2 = \mu_3$. H_1: at least one differs.

`SAY` **Factor B** — H_0: $\mu_1 = \mu_2$. H_1: they differ.

e Read the INTERACTION row first. Always.

`READ` **Effect Tests** table → the **A*B** row → Prob > F

A*B P-VALUE	WHAT YOU MUST DO
< 0.05	**Significant.** Main effects **cannot** be read separately. Compare all **cells**. Draw the circle diagram.
> 0.05	**Not significant.** Say so, then read the two main effect rows separately.

This single decision controls the whole rest of the problem.

f If the interaction IS significant

▼ **ON A*B LSMeans Differences Tukey HSD** → all cell combinations
▼ Factor Profiling › **Interaction Plots** → non-parallel lines confirm it

`SAY` Because the interaction is significant, the main effects cannot be interpreted separately. The effect of _____ on _____ **depends on** _____ .

g If the interaction is NOT significant

▼ **ON A FACTOR LSMeans Differences Tukey HSD** → that factor's levels

Only run Tukey on a **significant** main effect with **3+ levels**. A 2-level factor needs no Tukey — the F test already answered it; just report which level had the higher LSMean.

h The circle diagram, step by step

1 list every cell mean, **sorted high to low**
2 get the cutoff: HSD = q √(MSE/n) — JMP prints it
3 circle each run of means that stays **within** the cutoff
4 circles **overlap** — a mean can sit in two

Same thing as connecting letters: share a circle → not significantly different. The overlap is the point — X can match both the one above and the one below, while those two still differ from each other.

i Summary template

`SAY` A two-way ANOVA examined the effects of _____ and _____ on _____ . The interaction was **[significant / not significant]** (F = _____ , p = _____).

`SAY` *If significant:* the main effects cannot be interpreted separately; the effect of one factor depends on the other. Tukey comparisons of all cells showed _____ .

`SAY` *If not:* the main effects were interpreted separately. _____ was significant (F = _____ , p = _____) and Tukey showed _____ .

j Conditions for ANOVA models

▼ Row Diagnostics › Plot Residual by Predicted → shapeless cloud, no funnel
▼ Save Columns › Residuals → Distribution › Normal Quantile Plot

`SAY` **Independence:** the _____ units were randomly assigned, _____ to each of the _____ treatment combinations. **Normality:** the residual quantile plot shows points near the line. **Equal variances:** the residual by predicted plot shows even spread with no funnel.

Two-way checks **residuals**, not raw data. Cell SD ratio over 2 with only 4–5 per cell? Say the estimates are imprecise at that n and lean on the residual plot — don't claim a ratio you don't have.

WHERE EVERY NUMBER LIVES

YOU NEED	WHERE
group means, n	Means for Oneway Anova
group SDs	Tests that the Variances are Equal
equal-var p	Brown-Forsythe row
t, df, p pooled	**Pooled t Test** box
t, df, p Welch	**t Test** box
CI for difference	Upper / Lower CL Dif
paired mean diff	Mean Difference row
paired t, df, p	t-Ratio, DF, Prob > |t|
paired CI	Upper95% / Lower95%
one-way F, p	Analysis of Variance table
one-way pairwise	Compare Means › All Pairs, Tukey HSD
interaction test	**Effect Tests**, the A*B row
main effect tests	Effect Tests, single-factor rows
two-way pairwise	▼ on the effect › LSMeans Differences Tukey HSD
residual check	Row Diagnostics › Plot Residual by Predicted

FIVE TRAPS

1 · One-sided vs two-sided
"higher / greater / more" = one-sided → Prob > t or Prob < t. "a difference / differ" = two-sided → Prob > |t|. The right one-sided p is under 0.5.

2 · Forgetting Cross
Three effects in the box, and Model DF matches the formula. Otherwise your interaction never got tested.

3 · Reading main effects over a significant interaction
Interaction row first, every time.

4 · Wrong layout or modeling type
Fit Y by X = stacked. Matched Pairs = side by side. X must be Nominal.

5 · "All means are different"
H_1 for ANOVA is **at least one** mean differs.